

Mandla Ndhlovu

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Portfolio: mandla-portfolio.netlify.app | GitHub: github.com/MandlaCodes

Profile

AI Engineer specializing in machine learning, LLM systems, and cloud deployment. Skilled in building, fine-tuning, and deploying AI solutions using Python, TensorFlow, and AWS. Experienced in automating workflows, implementing RAG pipelines, and creating scalable, production-ready models with a strong focus on accuracy, optimization, and real-world impact.

Core Skills

Client Relationship Management | Workflow Automation | Neural Networks | Machine Learning | Predictive Analytics | AI Model Development | Python Programming | Data Visualization | LLM Systems | RAG Pipelines | LangChain | FastAPI | Django | AWS Deployment | Docker | CI/CD | Database Programming | Feature Engineering | Application Development | Auto Scaling & Load Balancing

Experience

MN Web Solutions | Full-Stack AI Engineer

Randburg, South Africa | Dec 2023 – Present

- Delivered custom AI solutions for clients, addressing unique business challenges and achieving measurable outcomes.
- Built and deployed production-ready AI/ML applications and multi-step reasoning workflows.
- Automated workflow processes, reducing repetitive tasks by 40%.
- Designed, trained, and deployed machine learning models to improve predictive analytics and client decision-making.

Avantgarde Landscaping | Part-Time Automation Specialist

Randburg, South Africa | Jan 2022 – Jan 2023

- Automated lead pipelines and invoicing from website submissions using Python scripts and AI tools.
- Streamlined business operations, saving 10–15 hours weekly.

- Built custom lead enrichment and scheduling tools to improve operational efficiency.
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Projects

FitAI – LLM Fitness Recommendation System

Developed an intelligent LLM-powered fitness assistant delivering personalized workout and nutrition plans.

- Integrated LangChain, LlamaIndex, and RAG pipelines to improve retrieval accuracy and context-aware recommendations and applied Prompt Engineering and Guardrails for safe, reliable LLM responses.
- Deployed via FastAPI backend and optimized inference with batch tokenization and model caching.
- Achieved a 25% improvement in recommendation accuracy and increased user engagement through adaptive feedback loops.

Mining Pillar Stability Prediction Tool

Collaborated with a Geologist from Exxaro to build a predictive ML system assessing coal pillar stability using LaModel and RS2 simulation data.

- Generated structured datasets from simulation outputs conducted by Geologist varying pillar geometry, depth, and geomechanical properties and built a full ML pipeline: data preprocessing, feature engineering, and normalization using Pandas and Scikit-learn.
 - Trained and optimized Random Forest and Random Regressor models to predict Extraction Safety Factor (ESF) with high accuracy and performed feature importance analysis to identify key parameters driving pillar stability, enabling safer stooping decisions.
 - Reduced analysis time by 80% and delivered actionable insights improving mining safety and operational efficiency.
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Education & Certifications

- Equivalent Experience: Full-Stack & AI/ML Engineering (Self-taught through applied projects and client work)
- AWS Solutions Architect – Associate
- Meta Back-End Developer
- DevOps Mastery Specialization

- Relational Database Administrator